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Late Paleozoic Paleomagnetic Results From the Junggar block, Northwestern China

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We have obtained middle Carboniferous to Permian paleomagnetic results from the Junggar block in order to test its tectonic coherence and to reconstruct its late Paleozoic position. From Hoboksar, on the northwestern margin of Junggar, a paleomagnetic pole from basic/silicic flows yields ϕ_p = 299°E, λ_p = 70°N (N = 15 sites, K/A_{95} = 36/ 6.4°). In the same general area, a pole from lava flows, plutons and as well as from remagnetized limestones/sandstones, is at ϕ_p = 350°E, λ_p = 69°N (N = 26 site, K/A_{95} =15/7.5°). Farther to the southwest in Keramayi, a pole from late Paleozoic dike swarms lies at ϕ_p = 230°E , λ_p = 80°N (N = 23 sites, K/A_{95} = 16/ 7.8°). In the southern margin near Urumqi, from sedimentary rocks and volcaniclastics of the Middle to Upper Carboniferous Qijiagou and Aoertu formations, a pole is produced at ϕ_p = 7°E, λ_p = 72°N (N = 15 sites, K/A_{95} = 28/7.4°). From the Middle Carboniferous Liushugou Formation and dike swarms intruding into the formation, two poles are located at ϕ_p = 231°E, λ_p = 69°N (N = 4 sites, K/A_{95} = 46/ 13.7°) and ϕ_p = 172°E, λ_p = 55°N (N = 7 sites, K/A_{95} = 17/ 15.1°), respectively, though reliabilities of these two poles are in question. Magnetic polarities of all the poles in our study are exclusively reversed. Based on positive fold tests and a baked contact test, we believe that these magnetizations were acquired during the Kiaman reversed interval (250-320 Ma). More accurate ages of the magnetizations cannot be resolved at present. Despite widely separate locations, the pole positions are roughly located on a small circle segment around the Junggar block. Similar inclinations suggest a coherent Junggar block since Permian time. Based on the results of this study and other published paleomagnetic data, a mean middle Carboniferous to Permian paleomagnetic pole is calculated at ϕ_p = 329°E, λ_p = 82°N (N = 6 poles, K/A_{95} = 26/13.4°). Recognition of late Paleozoic apparent polar wander for Junggar is obscured by a lack of detailed geochronologic data and possible relative horizontal rotations around vertical axis. The paleolatitude for the southern Junggar predicted by this pole is only 4° south of the present latitude. Difference exists between the mean pole from Junggar and Permian poles from other tectonic units in Asia (Tarim, Siberia, Kazakhstan, and North and South China), indicating post-Permian relative motions. At the boundary between Tarim and Junggar, this motion is likely to have a left-lateral sense.

INTRODUCTION

Central Asia is a complex mosaic of continental blocks, island arcs, and oceanic basins which are believed to have been assembled since late Paleozoic time. The Junggar block (physiographically, the Junggar basin) is located in northwestern China, covering an area of nearly 130,000 km^2 (Figure 1). The triangularly shaped Junggar basin is bounded by high ridges of the eastern Tian Shan on the south ("Shan" means mountain in Chinese). The Borohoro Shan, Tian Shan and Bogda Shan separate the Junggar basin from the Yili, Tarim, and Turpan basins. On the northwest, the basin is bounded by relatively low uplands of the West Junggar mountain system which are formed by folded Lower, Middle, and some Upper Paleozoic rocks. On the northeast, the basin is bounded by the Altai mountain system and the Kelameili uplift.

The origin and nature of the Junggar block have been controversial. In one hypothesis, the Junggar block is floored by Precambrian continental crust [*Zhang et al.*, 1984]. The main argument to support this hypothesis is that the depth of Moho ranges from 42 to 44 km [*Huang et al.*, 1987; *Ma*, 1987] as derived from gravity data. However, if the post-Devonian sediments are stripped away, the Junggar block is actually thinner than other continental blocks in central Asia such as Siberia [*Coleman*, 1989]. A second hypothesis suggests that the Junggar basin is underlain by early to middle Paleozoic oceanic crust trapped during the late Paleozoic consolidation of central Asia [*Feng et al.*, 1989; *Coleman*, 1989; *Carroll et al.*, 1990]. Highly deformed and dismembered ophiolites ranging in age from Cambrian to Devonian occur in fold belts on all sides of the Junggar block. Lead isotope studies on mid-Carboniferous (321 ± 7 Ma) anorogenic granitic plutons in the West Junggar Mountains suggest that they were derived from partial melting of oceanic crust with no contribution from Precambrian basement [*Feng et al.*, 1989; *Kwon et al.*, 1989]. Upper Paleozoic sandstones deposited in the eastern and southern Junggar basin are indicative of a magmatic arc provenance, further supporting the oceanic crust origin hypothesis [*Carroll et al.*, 1990]. *Hsu* [1988] proposed that the Junggar block is a Carboniferous back arc basin separated from Paleo-Tethys to the south by a late Paleozoic frontal arc. *Carroll et al.* [1990] concluded that extension of the Junggar Ocean, if it existed, must have ceased prior to the Permian based on sedimentary history records in the southern Junggar. *Watson et al.* [1987] and *Carroll et al.* [1990] interpreted Junggar as a flexurally subsiding foreland basin filled with non-marine clastics during the late Permian-early Triassic.

The southern margin of Eurasia in the late Paleozoic is believed to be marked by the Tian Shan-Hegen Suture near the northern boundary of China [*Zhang et al.*, 1984; *Huang et al.*, 1987]. This view is supported by the analysis of the Visean coral faunas and the Cathaysian, Angara, and Kuznetsk floras, which suggests that the discontinuity between the Cathaysian and the Middle Asian coral faunas and as well as the boundary between the Cathaysian and Angara/Kuznetsk floras coincide with the Tian Shan fold belt [*Hu*, 1985; *Smith*, 1988]. If so, the Junggar block must have occupied a higher latitude region than the Tarim, North China, and South China blocks during Carboniferous and Permian.

Paleomagnetic data can help to measure displacement histories of individual terranes which make up the Junggar block. Prior to our study, there were few published paleomagnetic results from the Junggar block [*Li et al.*, 1989; *Chen et al.*, 1991]. In 1988, we collected late Paleozoic rocks

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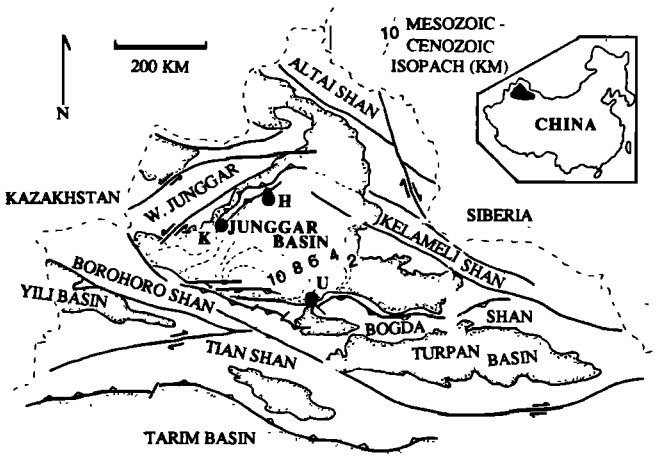


Fig. 1. Generalized tectonic map of northwestern China. Circles indicate the paleomagnetic sampling region: H, Hoboksar; K, Keramayi; and U, Urumqi (modified from *Carroll et al.* [1990]).

in three regions along the margins of Junggar under a cooperative paleomagnetic research project between the Geophysics Department at Stanford University and the Xinjiang Geology Institute in Urumqi. The areas sampled for paleomagnetic study are Hoboksar and Keramayi on the northwestern margin and Urumqi on the southern margin of Junggar (Figure 1). The results from Late Paleozoic dike swarms from the Keramayi region have been published elsewhere [*Li et al.*, 1989].

GEOLOGY

The stratigraphy of the Junggar block can be divided into pre-Carboniferous, Carboniferous, Permian volcanic and sedimentary rocks, and Mesozoic and Cenozoic nonmarine sedimentary rocks [*Xinjiang Stratigraphic Compiling Group*, 1981; *Feng et al.*, 1989; *Coleman*, 1989; *Lee*, 1985; *Ulmishek*, 1986]. The pre-Carboniferous basement of the basin is composed of Ordovician, Silurian, and Devonian sedimentary and metamorphic complexes, which are well exposed in the eastern and western margins of the basin. The Lower Carboniferous rocks are dominantly volcanic, overlain by Upper Carboniferous marine flysch with only a few scattered volcanic rocks including ash layers. Marine sedimentation continued in the central and southwestern parts of the basin. But by late Permian time, the marine seaway was completely closed, giving way to continental intermontane-sedimentary deposition. Mesozoic and Cenozoic continental fluvial and lacustrine deposits of coarse to fine clastics are very thick in the southern parts of the basin (Figure 1).

The Upper Paleozoic Qijiagou section is located 20 km southeast of Urumqi (Figure 1), near Qijiagou village. The section is on the southern edge of the Junggar basin and the northern foothills of the Bogda Shan. Structurally, it is on the north limb of a large east-west trending Neogene faulted anticlinorium. The oldest rocks in the Bogda Shan are a 2-3 km thick section of Lower to Middle Carboniferous andesitic volcaniclastics. The basinal deposits are overlain by an 800-m regressive marine sequence, which at its base consists of carbonate and volcaniclastic turbidites. Productid brachiopods from the Qijiagou Formation have been dated as Middle to Upper Carboniferous [*Carroll et al.*, 1990]. The turbidite section grades upward into Upper Carboniferous slope and shelf shales and volcaniclastic sandstones. Stromatolites deposited in the early Permian record the final stages of withdrawal of the Junggar ocean.

We have sampled three Middle-Upper Carboniferous formations: Liushugou (C_{2l}), Qijiagou ($C_{2\text{-}3q}$), and Aoertu (C_{3a}). The Liushugou Formation is the oldest and composed mainly of andesitic volcanic breccia, tuffaceous breccia, tuff intercalated with andesite porphyrite and minor siltstone and limestone lenses, bearing abundant fusulinids [*Liao et al.*, 1987]. We collected oriented samples at 13 sites (8uc3 to 8uc15) from the upper part of the formation, seven of which were from dike swarms intruding into the section. The Qijiagou Formation has a total thickness of 281 m and is divided into three members according to lithological character. It mainly consists of volcaniclastics, sandstones interbedded with sandy limestone and biocalcilutite. The contact between the Qijiagou and Liushugou formations is conformable. The upper member of the Qijiagou Formation from which our seven sites (8uc1,2, 24-28) were collected is 189 m thick, containing more thickly bedded limestone than the other members. The lower part of this member is yellow-green, thin-bedded siltstone and fine-grained sandstone interbedded with sandy limestone. Abundant fossils, including fusulinids, foraminifera, corals, brachiopods, are found in the formation, yielding a middle to late Carboniferous age [*Liao et al.*, 1987]. The Upper Carboniferous Aoertu Formation is 175 m thick and is mainly composed of dark green, yellow-green, and gray-black siltstone and sandstone. In the lower part, it is intercalated with a small amount of thin-bedded and lensoid sandy limestone, decreasing gradually upward. The upper part is dominated by gray to black siltstone intercalated with pebble-bearing sandstone and conglomerate. Eight sites (8uc16-21) were collected from grayish black siltstone of the upper part of the Aoertu Formation.

Hoboksar is located in the northwest part of the Junggar basin (Figure 1). The oldest rocks exposed are Middle-Upper Ordovician, in turn overlain by Silurian, Devonian, and Carboniferous marine deposits.

The Upper Devonian Honggulerun Formation (D_{3h}) consists of a 595-m section of siliceous siltstone, sandstone, and muddy limestone, containing corals *Nalivkinella* sp., *Dictyoclostus* sp., *Cyrtospirifer* sp., *Athyris sulcifer Nalivkin*, *Platyrachella* sp., *Camarotoechia* sp., *Plicatifera* sp., *Productella* sp., and trilobite *Phacops* sp. (Xinjiang Bureau of Geology, unpublished report, 1973). It is conformable with both underlying and overlying strata. Three sites (8hp5-7) were collected from a 20-m overturned section of thin bedding limestones in Bulungou. The Heshantou Formation (C_{1h}) consists of gray siliceous mudstone, siltstone, calcareous sandstone, and calcareous limestone. Brachiopods in the limestones indicate a Visean age (C_1). The upper part of the formation grades into conglomerate layers and contains thin coal beds. The Heshantou Formation is 1500 m thick and is conformable with both the overlying Carboniferous Sarlebulake Formation and underlying Upper Devonian Talebahatai Formation. Three sites (8hc8-10) were collected from the Heshantou Formation in a thin limestone section in Heshantou. The Sarlebulake Formation (C_{1s}) consists of 592 m of gray-black, yellowish sandstone, conglomerate, and siltstone, interbedded with extrusive and volcaniclastic rocks. Fossils found in the formation are *Angaropteridium cardiopteroides* (*Sehmal.*) *Zalessky*, *Calamites* sp., and *Neoggerathiopsis* sp. There is a hiatus between the Sarlebulake and overlying Harlegiawu formations. Three sites (8hp16-19), in addition to one from a pluton intruding into the Sarlebulake Formation, were collected from the Sarlebulake Formation.

The Harlegiawu Formation (C_{2h}) is a 234-m suite of continental silicic volcanic and volcaniclastic rocks, containing plant fossils: *Calamites* sp. and *Noeggerathiopsis* sp. The Karlargang Formation (C_{2k}) consists of 1436 m rhyolitic flows, tuff, and interbedded sandstone and shale layers. The lower part of formation contains coal layers with plant fossils *Angaropteridium cardiopteroides (Schmal) Zalessky*, *Cardioneura sp.*, *Sphenophyllum* sp., *Pursongia* sp., *Noeggerathiopsis* sp., etc. The Karlargang Formation lies conformably upon the Harlegiawu Formation but is overlain unconformably by Upper Jurassic sandstone. The ages of the Harlegiawu and Karlargang formations are assigned to the middle Carboniferous (Xinjiang Bureau of Geology, unpublished report, 1973), based on regional lithologic comparison. A total of 18 sites were collected from the Karlargang and Harlegiawu formations.

U-Pb, Rb-Sr and ^{40}Ar-^{39}Ar isotopic ages on plutons and volcanic rocks in western Junggar cluster around 325 Ma [*Bureau of Geology and Mineral Resources of Xinjiang*, 1987].

However, isotopic age data are sparse in the region we have studied. Two K-Ar ages from granitic plutons in the Hoboksar region are 278 and 267 Ma; one age from volcanic rocks is 243 Ma.

PALEOMAGNETIC STUDY

General Procedure

Generally, we collected six to eight field-drilled samples at each site using a portable drill. Samples were oriented in situ using a Brunton Compass and in many cases were checked by a Sun shadow angle reading. No significant difference was found between magnetic and solar azimuths.

Most samples were measured and demagnetized in the paleomagnetic laboratory at Stanford University using an automated cryogenic magnetometer with integrated, alternating field (AF) demagnetization coils and a Schonsted (TSD-1) oven. For specimens subjected to thermal demagnetization, bulk magnetic susceptibility as a function of heating step was monitored using a Bartington magnetic susceptibility meter, model M.S.2. A few samples were measured and demagnetized by thermal and AF methods in the paleomagnetic laboratory of the University of New Mexico.

AF demagnetization treatments consisted of 8-15 steps in progressively higher fields up to a peak field of 80 mT. In thermal demagnetization treatments, samples were heated in 50°C or 100°C steps from 50°C to a maximum of 690°C depending upon the rock type. After each thermal demagnetization step, a 5-mT AF demagnetization was performed just before measurement inside the magnetometer to eliminate viscous components acquired while the sample was transferred from the oven to the magnetometer. AF or thermal demagnetization results are plotted on orthogonal vector diagrams [*Zijderveld*, 1967]. Principal component analysis [*Kirschvink*, 1980] was used to obtain a characteristic direction and in some cases a lower H_C/T_{lub} (H_C: coercive force; T_{lub}: laboratory unblocking temperature) component for each specimen.

Urumqi Region

Liushugou Formation (C_{2l}) and associated dikes. Thirteen sites from the Liushugou Formation were collected from a continuous section in Liushugou (the Liushu Valley, "gou" means valley in Chinese). Six sites come from andesitic flows, tuffs, and interbedded clastics. Seven sites come from mafic dikes intruding into the bedded rocks. In many cases, the dikes were intruded into the section along existing fractures or faults. The age of the dikes is not known, although we have not seen any dike intruding into the overlying Qijiagou and Aoertu formations. Samples in most sites show well-grouped and stable magnetizations after AF or thermal demagnetization, although a presumably viscous magnetization is usually removed during the low range of AF or thermal demagnetization. Typical orthogonal demagnetization diagrams are shown in Figure 2. The directions of magnetization of individual specimens from site 8uc15 are not well defined and are inconsistent within the site. The mean direction from site 8uc11 is not consistent with the overall population (Table 1). Thus data from sites 8uc11 and 8uc15 are not used for the final calculation.

The mean direction from the Liushugou Formation is significantly different from that of the dikes which intruded it, and therefore they are not combined (Figure 3 and Table 1). The difference suggests that the magnetizations were acquired at different times. The mean inclination from the dikes is not significantly different from the mean inclination of the Qijiagou and Aoertu formations (described below). The difference in declination is significant, which could indicate that the Junggar block experienced a significant orientation shift (some 70°) between the middle Carboniferous and late Permian. Since only seven dikes were sampled and tilt correction is made on the assumption that the dikes intruded flat beddings, the difference may reflect insufficient averaging of geomagnetic secular variation and inaccurate tilt corrections.

Tilt-corrected directions of the dikes are better grouped than those in situ, while for the lava flows/volcaniclastics of the Liushugou Formation, the situation is reversed (Figure 3). In both cases, the change of grouping is not statistically significant. Because of the small number of sites and the divergence of the directions after tilt correction, we have chosen not to use the results from the Liushugou Formation for tectonic interpretation.

Qijiagou (C_{2-3q}) and Aoertu (C_{3a}) formations. Samples of the Middle to Upper Carboniferous Qijiagou Formation were collected from Qijiagou and Serenzhigou. Sites 8uc24 to 8uc26 were collected from a coherent section dipping to the northwest. Sites 8uc24 and 8uc25 come from basaltic flows below and above a 15-m-thick limestone lens. Site 8uc26 is from pillow basalts, about 20 m stratigraphically above site 8uc25. The pillows are not widely exposed and probably reflect a small, isolated fissure eruption. Sites 8uc27 and 8uc28 were collected from thickly bedded dark gray sandstones.

The Upper Carboniferous Aoertu Formation has been sampled at two localities. Sites 8uc16-17 were collected from a roadcut in Gejiagou, and sites 8uc18-23 were collected from a 100-m coherent section in Jinjinzhigou. At Jinjinzhigou, the samples were collected from a monocline with a large variation in bedding attitudes so that a fold test can be attempted. The demagnetization behavior of samples from the Qijiagou and Aoertu formations is straightforward (Figure 2). Except for site 8uc2, well-grouped directions at each site are readily isolated. Site 8uc2 is located in volcaniclastic rocks, and only two samples from black tuffaceous siltstone show a direction of magnetization consistent with that of site 8uc1, a few meters below stratigraphically. Other samples from gray coarse-grained sandstone exhibit directions scattered around the present Earth field (PEF) direction. Only two specimens from site 8uc26 are measurable; others have been damaged before arrival at the laboratory. AF demagnetization was most effective in removing viscous components parallel to the PEF direction (Figure 2).

The mean direction of the Qijiagou Formation is indistinguishable from that of the Aoertu Formation, suggesting either that the polar wander of the Junggar block was negligible from late middle Carboniferous to late Carboniferous times, the Qijiagou Formation was remagnetized at about the same time the Aoertu Formation acquired a primary magnetization, or both the Qijiagou and Aoertu formations were remagnetized at some later time. Because of their similarity, the two groups of directions were combined together to yield a mean direction.

Variable bedding attitudes allowed us to conduct a fold test on the Qijiagou and Aoertu formations. Tilt-corrected directions are statistically better grouped than the in situ directions at the 99% confidence level, establishing a prefolding age of the magnetization (Figure 3 and Table 1), though the last major phase of tectonic activity in the region occurred in the Neogene. The positive fold test, together with exclusively reversed polarity, is consistent with the hypothesis that the magnetization was acquired during the Kiaman reversed polarity interval (320-250 Ma)[*Harland et al.*, 1982].

Hoboksar Region

Samples from Hoboksar were collected from two locations: Heshantou (47°11'N, 86°41'E) and Bulungou (46°42'N, 86° 01'E).

The Harlegiawu and Karlargang formations (C_{2h} and C_{2k}). We collected samples of silicic volcanic rocks from the Harlegiawu and Karlargang formations, near Bulungou village, in both limbs of a syncline which trends in approximately NNW-SSE. From the western limb which dips to the northeast, sites 9hp37-45 were collected; sites 8hp46-54 were collected from the eastern limb dipping to the northwest. The 18 sites are distributed along directions perpendicular to the strikes of bedding. Bedding attitudes were measured from interbedded sedimentary beds or tuff layers. The dips in the western limb of the syncline range from 11° to 17° and 20° to 40° in the

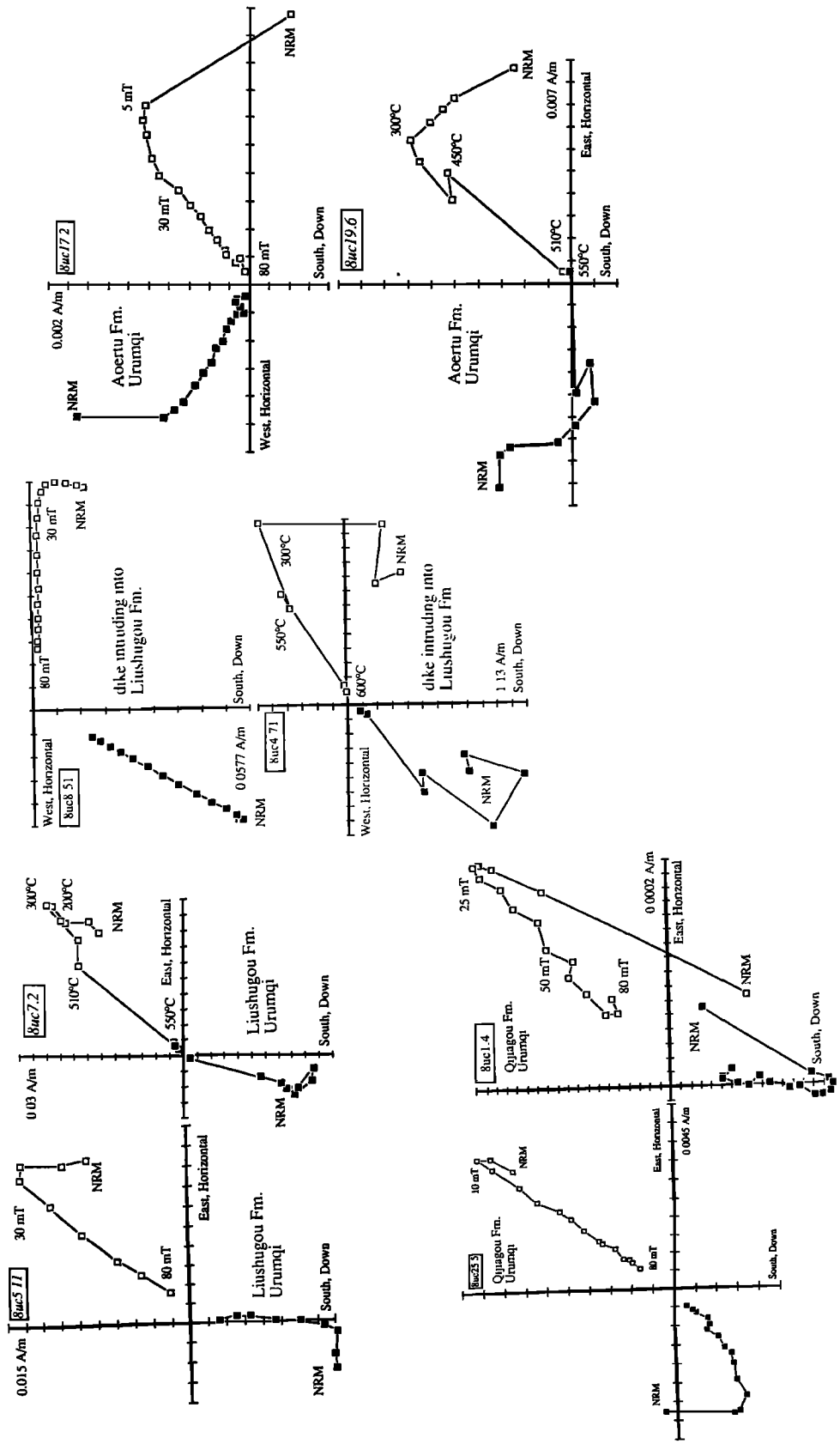


Fig. 2. Orthogonal vector thermal and AF demagnetization diagrams of specimens from the Urumqi region. Solid symbols refer to the projection on the horizontal plane; Open symbols refer to the projection of inclination (geographic coordinates).

TABLE 1. Urumqi C_{2-3} (Latitude 43°47'N, Longitude 87°45'E)

Site*	Age	N(M)	D	I	D_c	I_c	k	a_{95}	L	S(D)
Aoertu Formation (Late Carboniferous, C_{3a}) and Qijiagou Formation (Middle Carboniferous, C2q)										
8uc16	C_{3a}	6(6)	291	-51	175	-56	29	12.6	SS	231(64W)
8uc17	C_{3a}	5(5)	297	-50	163	-63	74	7.8	SS	226(63W)
8uc18	C_{3a}	6(6)	282	-43	152	-65	62	8.6	SS	210(67W)
8uc19	C_{3a}	6(6)	280	-44	153	-63	51	9.5	SS	210(67W)
8uc23	C_{3a}	6(6)	240	-59	174	-60	110	6.4	SS	206(35W)
8uc22	C_{3a}	6(6)	244	-65	157	-62	67	8.2	SS	204(38W)
8uc20	C_{3a}	6(6)	251	-67	148	-63	299	3.9	SS	203(40W)
8uc21	C_{3a}	6(6)	267	-69	137	-67	655	2.6	SS	203(40W)
8uc28	C2q	6(6)	214	-64	147	-49	55	9.1	SS	197(42W)
8uc27	C2q	6(6)	230	-63	150	-56	47	8.9	SS	197(42W)
8uc26	C2q	2(2)	196	-67	139	-43	-	-	BS	197(42W)
8uc25	C2q	6(6)	233	-59	157	-57	46	9.9	BS	197(42W)
8uc24	C2q	5(6)	256	-46	191	-69	15	20.0	BS	197(42W)
8uc2	C2q	2(5)	213	-40	165	-64	-	-	VC	158(41W)
8uc1	C2q	5(5)	209	-53	113	-76	31	14	LS	140(42W)
$A+Q_b$	C_{2-3}	15	250	-60	-	-	17	9.6		
$A+Q_a$	C_{2-3}	15	-	-	155	-62	52	9.7		
Liushugou Formation (Middle Carboniferous)										
8uc5	C_{21}	6(6)	182	-48	186	-67	53	9.2	VC	86(19S)
8uc9	C_{21}	6(6)	203	-27	222	-65	23	14.0	VC	97(41S)
8uc7	C_{21}	6(6)	188	-45	188	-79	118	6.2	VC	98(34S)
8uc11†	C_{21}	6(6)	193	-78	64	-58	25	13.7	FL	140(41S)
8uc12	C_{21}	4(5)	202	-51	123	-72	41	14.6	VC	140(41S)
8uc15†	C_{21}	3(4)	228	30	228	-18	23	26.6	VC	118(50S)
L_b	C_2	4(6)	194	-43	-	-	38	15	-	-
L_a	C_2	4(6)	-	-	185	-74	32	16	-	-
Dikes Intruding Into Liushugou Formation										
8uc3	C_{21}	6(6)	200	-21	209	-56	568	2.8	DK	97(36S)
8uc4	C_{21}	7(7)	227	-7	247	-42	31	10.9	DK	97(52S)
8uc6	C_{21}	6(6)	195	-37	203	-55	24	13.8	DK	86(19S)
8uc10	C_{21}	6(6)	213	-9	224	-44	50	9.6	DK	97(41S)
8uc8	C_{21}	5(6)	207	-3	221	-51	150	6.3	DK	97(53S)
8uc13	C_{21}	7(7)	215	-34	266	-74	19	14.0	DK	107(49S)
8uc14	C_{21}	4(4)	213	-12	227	-58	36	15.5	DK	107(49S)
$Dikes_b$	C_2-P	7(7)	210	-18	-	-	23.8	12.6	-	-
$Dikes_a$	C_2-P	7(7)	-	-	226	-56	28.7	11.5	-	-

N(M), number of samples used (measured) or number of sites in the grand mean; *D*, *I* and D_c, I_c are declination and inclination in geographic and stratigraphic coordinates. *k* and a_{95}, precision parameter estimate and radius of the 95% confidence limit in degrees. C_2, C_3, and C_{2-3} indicate middle Carboniferous, late Carboniferous and middle-late Carboniferous. L, lithological description, LS, limestone; SS, sandstone; VC, volcaniclastic; DK, dike; FL, flow. S(D): strike and dip directions.

*, $A + Q_b$ and $A + Q_a$ are the in situ and tilt-corrected grand mean directions of the Qijiagou and Aoertu formations. L_b and L_a are the in situ and tilt-corrected grand mean directions of the Liushugou Formation. $Dikes_b$ and $Dikes_a$ are the in situ and tilt-corrected grand mean directions of the dikes intruding into the Liushugou Formation.

†, Excluded from the grand mean. The grand mean direction is a Fisherian average of site mean directions.

eastern limb. As discussed below, the syncline probably formed in late Paleozoic time and certainly was not formed later than the Jurassic because the Karlargang Formation is overlain unconformably by Jurassic sandstones. Most of samples are silicic volcanics, but sites 8hp44-47 come from andesitic basaltic flows which lie close to a fault along the foothill. As a result, bedding attitudes of sites 8hp44-46 are obscured.

The samples can be split into two groups based upon their behavior during AF and thermal demagnetizations. Samples from the first group have moderate coercivities, and the characteristic component of magnetization can be isolated after AF treatments of 60-80 mT. Thermal demagnetization is as effective as AF demagnetization in removing occasional viscous components. Magnetic components of the first group can be selected according to end-points or linear segments to the origin. The first group includes sites 8hp37-39, 8hp43-44, 8hp46-50, and 8hp52-54 (Figure 4). The mean direction of these sites after tilt correction is D/I =166°/-49°, N = 11 sites, R =10.75, k = 40 and a_{95} = 7.3°, with no correction for plunge. This mean direction is not significantly different from the mean direction of all sites and thus in our analysis we have used the overall mean direction (Table 2).

The samples from the second group cannot be demagnetized by AF treatment since they have coercivities higher than 80 mT, which is the highest peak AF field we can attain with our instruments. During thermal demagnetization, characteristic components similar to these isolated from the first group are found. But the directions are not well defined as linear segments. One noticeable difference between these two groups is different magnetic phases as shown by unblocking temperature. For the first group, magnetite is the main magnetic phase for the characteristic components, whereas hematite is the dominant magnetic phase for the samples in the second group. Results of sites 8hp44-46 are not included in calculating the grand mean direction as bedding attitudes are not clear for these sites.

Based upon measurements of bedding attitude from the west and east limbs of the syncline, a hinge line of azimuth (43°)

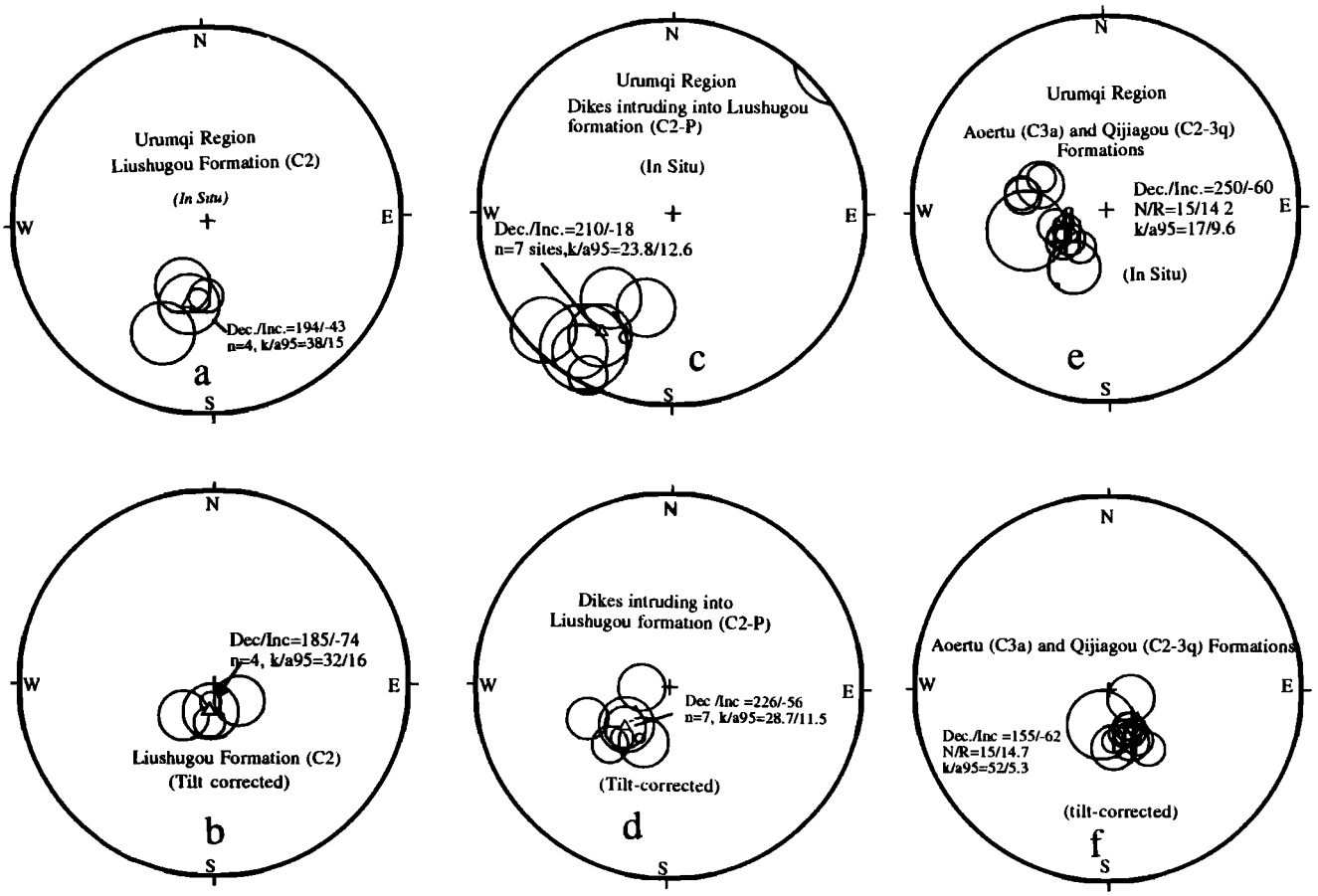


Fig. 3. Equal-angle stereographic projections of site-mean directions of the Liushugou Formation (C_{2l}, Figures 3*a* and 3*b*), dikes intruding into the Liushugou Formation (Figures 3*c* and 3*d*), and the Qijiagou and Aoertu formations ($C_{2\text{-}3q}$ and C_{3a}, Figures 3*e* and 3*f*). Circles are the 95% confidence limits around the site-mean directions (centers). Average of site-mean direction is plotted as open triangle with the 95% confidence limit. All projections are on upper hemispheres.

and plunge (13°) is established by a great circle fit of poles to bedding planes. Site mean directions of magnetization were first rotated (along with poles to bedding plane) to correct for plunge, then rotated to correct for tilt to yield a grand mean direction (Table 2). For comparison, site-mean directions and grand mean direction based on convention tilt correction (one-step) are also listed in Table 2. The net difference between two means calculated by different correction procedures is only 2° in declination because of the shallow 13° plunge.

The tilt-corrected directions are better grouped than those before tilt corrections at p =0.01 (Figure 5) [*McElhinny*, 1964; *McFadden and Jones*, 1981], establishing a pre-Jurassic magnetization age. The exclusively reversed polarity constrains the age of the magnetization to be middle-late Carboniferous to Permian, corresponding to the Kiaman reversed polarity interval.

A low H_c/T_{lub} normal polarity component usually removed in the range of less than 20 mT or 200° can be isolated from almost all sites. The mean direction of this component for a total of 18 sites is D/I =4°/63°, k = 58 and a_{95} = 4.6° in geographic coordinates, indistinguishable from the present Earth field direction at the sampling location. Scatter of this component increases when corrected for tilt, giving a mean direction at D/I =1°/46°, k =18 and a_{95}=8.5°. We believe it to be a Quaternary viscous component.

Honggulerun Formation (D_{3h}). Small viscous components removed during demagnetization are roughly aligned along the PEF direction. Both AF and thermal demagnetizations reveal a characteristic component with a southeast declination and a moderate positive inclination. Demagnetization trajectories in the high thermal or AF demagnetization ranges converge toward the origin (Figure 4). The mean of the tilt-corrected characteristic components is not statistically different from the mean directions of the younger Heshantou and Sarlebulake formations, as described below.

Because late Paleozoic plutons and volcanic rocks are exposed in the vicinity of the sampling locality, we believe that the characteristic magnetization isolated from the Devonian Honggulerun Formation may represent an overprint caused by plutons or volcanic rocks during late Paleozoic but before the tilting. This view is supported by (1) the observation that the mean direction in stratigraphic coordinates is not significantly different from those of the Carboniferous Heshantou Formation (C_{1h}) and Sarlebulake Formation (C_{1s}) (Table 3) and (2) the magnetic polarity is exclusively reversed.

The Heshantou Formation (C_{1h}). The magnetic behavior of samples from the Heshantou Formation (C_{1h}) becomes univectorial to the origin after demagnetization to 20 mT or 200° C (Figure 4). These characteristic components have northwest declinations and negative inclinations, far removed from the in situ present Earth field direction. However, after tilt correction, the characteristic components become aligned with the stable magnetic directions isolated from other the Upper Paleozoic formations.

The Sarlebulake Formation (C_{1s}) and baked contact test. Similar in situ directions from sandstones of the Lower Carboniferous Sarlebulake Formation and a pluton intruding into the sandstones (sites 16-18) suggest that the sandstones near the contact carry a secondary magnetization (Figure 6). Tilt-corrected directions of magnetization of site 8hc19 from the same formation but some 2 km away from the pluton are indistinguishable from the in situ direction of sites 8hp16-18 (Figure 6 and Table 3) and the tilt-corrected directions of the

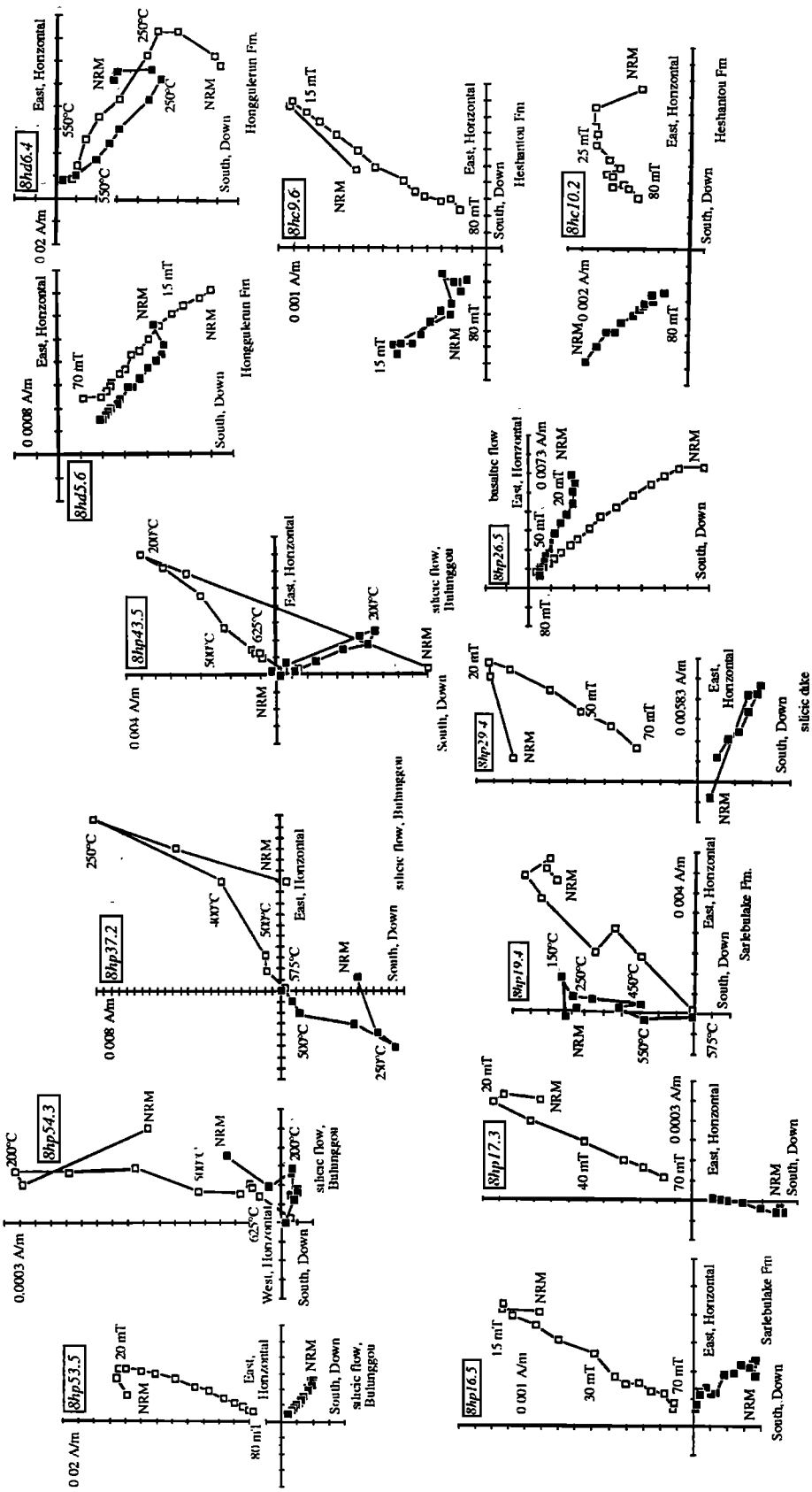


Fig. 4. Orthogonal vector thermal and AF demagnetization diagrams of specimens from the Hobuksar region. Solid symbols refer to the projection on the horizontal plane; Open symbols refer to the projection of inclination (geographic coordinates).

TABLE 2. Hoboksar, Silicic Flows C_{2-3}-P_1 (Latitude 46°42'N, Longitude 86°03'E)

Site	Age	N(M)	D	I	D_c	I_c	k	a_{95}	L	S(D)
					West Limb					
8hp37	C_{2k}	5(5)	202	-54	188	-46	50	10.9	RF	224(15W)
8hp38	C_{2k}	4(5)	181	-57	193	-50	55	12.5	RF	330(11E)
8hp39	C_{2k}	5(6)	164	-66	190	-57	63	9.7	RF	325(17E)
8hp40	C_{2k}	3(6)	148	-47	165	-45	52	17.2	RF	331(17E)
8hp41	C_{2k}	6(6)	156	-43	168	-37	97	6.8	RF	320(15E)
8hp42	C_{2k}	6(6)	157	-42	170	-41	59	12	RF	340(15E)
8hp43	C_{2k}	6(6)	168	-47	181	-45	47	9.9	SS	348(13E)
M37-43b	C_2	7(7)	167	-52	-	-	33	10.7	-	-
M37-43a	C_2	7(7)	-	-	178	-46	67	7.4	-	-
M37-43ap	C_2	7(7)	-	-	178	-46	63	7.6	-	-
					East Limb					
8hp47	C_{2h}	6(6)	150	-61	161	-42	112	6.3	AF	267(20W)
8hp48	C_{2h}	6(6)	190	-81	163	-53	188	4.9	AF	240(30W)
8hp49	C_{2k}	5(6)	137	-85	139	-51	154	6.2	AF	225(34W)
8hp50	C_{2k}	6(6)	201	-88	160	-49	287	3.9	RF	244(39W)
8hp51	C_{2k}	6(6)	313	-85	161	-56	27	13.2	RF	244(39W)
8hp52	C_{2k}	6(6)	210	-84	167	-51	367	3 5	RF	245(35W)
8hp53	C_{2k}	6(6)	140	-72	161	-39	134	5.8	RF	258(35W)
8hp54	C_{2k}	6(6)	111	-77	153	-49	149	5.5	RF	256(33W)
M47-54b	C_2	8(8)	151	-82	-	-	50	7.9	-	-
M47-54a	C_2	8(8)	-	-	155	-49	104	5.5	-	-
M47-54ap	C_2	8(8)	-	-	158	-49	105	5.4	-	-
					Mean of Both West and East Limbs					
Tb	C_2	15(15)	163	-68	-	-	17	9.5	-	-
Ta	C_2	15(15)	-	-	166	-48	46	5.7	-	-
Tap	C_2	15(15)	-	-	168	-48	52	5.3	-	-
Ib	C_2	15(15)	-	-66	-	-	12	13	-	-
Ia	C_2	15(15)	-	-	-	-47	99	4.4	-	-
					Sites Excluded From the Mean					
8hp44	C_{2k}	4(6)	186	-52	-	-	72	10.9	BF	-
8hp45	C_{2k}	7(7)	174	-47	-	-	138	5.0	BF	-
8hp46	C_{2k}	5(5)	205	-47	-	-	115	7.2	AF	-

Abbreviations are the same as in Table 1. For lithology, RF, rhyolitic flow; SS, sandstones; BF, basaltic flow; AF, andesitic flow. M37-43b and M37-43a, and M47-54b and M47-54a are in situ and tilt-corrected grand mean directions of sites 37-43 (west limb), and sites 47-54 (east limb). Tb and Ta are in situ and tilt-corrected grand mean directions of sites 37-43 and sites 47-54 all together. M37-43ap, M47-54ap and Tap are the final grand mean directions corrected for plunge (dip vector: 43°, dip: 13°NE). Ib and Ia are mean inclinations before and after tilt correction based on inclination-only average.

other sedimentary units. Assuming that both localities were tilted at the same time, this observation suggests that (1) the characteristic magnetization of the site far removed from the pluton has not been overprinted by the pluton; (2) tilting of the formation occurred before the intrusion of the plutons; and (3) the age of the pluton is younger than early Carboniferous but within the Kiaman interval (250-320 Ma).

Sites in plutons, dikes, and lava flows and tilt test. Sites 8hd1 and 2 were collected in a roadcut just south of the Hoboksar township. Sites 8hd3-4, and 8hp35-36 are from intrusions intruding into the Honggulerun (D_{3h}) Formation near Bulungou. Sites 8hc11-13 are located in plutons at Heshantou. Sites 8hp14 and 15 were collected from a dike and a lava flow in roadcuts near Heshantou. Bedding attitudes of sites 8hp20-26 were determined by averaging several readings of lava flow surfaces. Sites 8hp27-30 and sites 8hp32-34 were collected from silicic dikes in the vicinity of site 8hp31, which is collected from sandstones of the Sarlebulake Formation(?). Stable, consistent magnetizations could not be obtained from sites 8hc12-13, 8hp25, and 8hp34. Sites 8hp33 and 8hp35 are badly weathered and give directions which are not consistent with directions isolated from the other sites at the same localities. Site 8hp28 probably records a portion of transitional behavior of the geomagnetic field. The paleohorizontal control on sites 8hd1-2, and 8hc11 are uncertain. Thus results from sites 8hc12-13, 8hp25, 8hp33-35, 8hd1-2, and 8hc11 bear no tectonic significance and will not be discussed further. All samples in other sites demonstrate stable magnetic directions (Figure 4 and Table 3). These sites are either plutons/lava flows in contact with sedimentary rocks, or dikes intruded vertically.

The grand mean direction of the sites in plutons, lava flows and dikes is similar to the mean directions of the Honggulerun (D_{3h}), Heshantou (C_{1h}), and Sarlebulake (C_{1s}) formations, indicating that either (1) magnetizations in different formations and different types of rock were acquired essentially at about the same time or (2) there is little apparent polar wander (APW) during the Carboniferous for the Junggar block. Because all the sites are reversely magnetized, we believe that age of magnetization may be constrained within the Kiaman reversed polarity interval (250-360 Ma). Because of the secondary nature of magnetizations in the sedimentary units we have combined all data from the plutons, lava flows, dikes, and sedimentary units together to produce a mean direction from Hoboksar.

Remagnetization of the sedimentary units (D_{3h}, C_{1h}, and C_{1s}) must have occurred before folding because the tilt-corrected directions are significantly better grouped than the in situ directions (Table 3). Similarly, if we perform a tilt test on the tilted lava flows, plutons, and sedimentary rocks, a

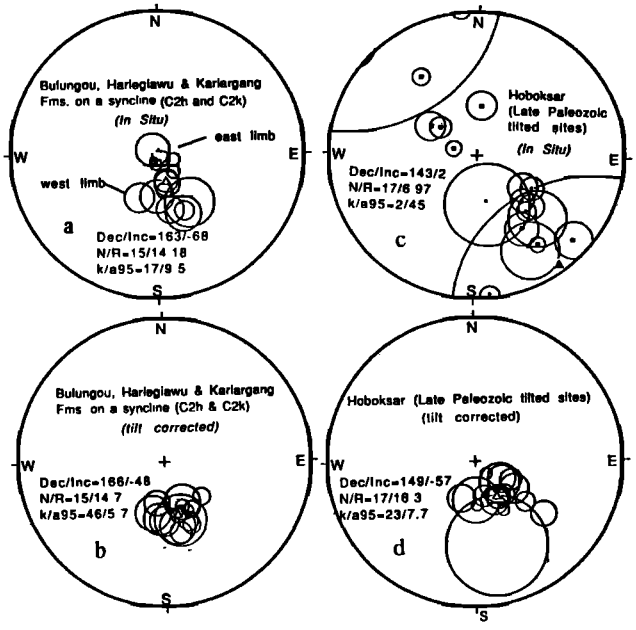


Fig. 5. Equal-angle stereographic projections of site-mean directions of the Harlegiawu and Karlargang formations (C_{2l} and C_{2k}, Figures 5*a* and 5*b*, all projected on upper hemispheres); remagnetized sedimentary units, tilted plutons and lava flows (Figures 5*c* and 5*d*, solid dots are projected on lower hemispheres, open squares are projected on upper hemispheres). Circles are the 95% confidence limit around the site-mean directions (centers). Average of site-mean direction is plotted as open or solid triangle (upper or lower hemisphere projection) with the 95% confidence limit.

positive tilt test is obtained at $p = 0.01$ (Figure 5 and Table 3). The in situ directions of some dikes and pluton (sites 8hp14, 8hp18, 8hp27, and 8hp29-32) in Hoboksar are indistinguishable from the tilt-corrected directions of the sedimentary units (Figure 5 and Table 3). This fact indicates that igneous activity and folding in the Hoboksar region were concurrent. This observation leads us to believe that the syncline within the Harlegiawu and Karlargang formations formed shortly after the eruptions of silicic flows and hence conclude that the magnetizations of these formations (C_{2h} and C_{2k}) are primary.

Isothermal Remanent Magnetization Acquisition Experiments

Isothermal remanent magnetization (IRM) acquisition experiments on some representative samples are shown in Figure 7. Results of all samples, except 8hp54.4 from a rhyolitic flow and 8hp29.6 from a pluton which are not saturated before 1 T, indicate the absence of appreciable high coercivity magnetic phases.

Discussion and Tectonic Interpretation

Each site-mean direction was converted into a virtual geomagnetic pole (VGP). Then, paleomagnetic pole positions for formations/locations were calculated assuming the VGPs belong to a Fisherian statistical distribution [*Fisher*, 1953]. Late Paleozoic paleomagnetic poles reported from different regions of Junggar are listed in Table 4, and some of them have been plotted in Figure 8. In addition to the poles obtained from this study, they also include a middle-late Permian pole from the Urumqi region (JG6 of Table 4 [*Sharps*, 1990]), a late Permian pole from the Heavenly Lake region, southern Junggar (JG5 of Table 4 [*Nie et al.*, 1989]), a remagnetized Permian pole from the Keramayi region, western Junggar (JG 9 of Table 4 [*Zhao et al.*, 1990]), and a late Paleozoic pole from the dike swarms in the Keramayi region (JG 3 of Table 4 [*Li et al.*, 1989]).

The pole JG6 was calculated from a synfold magnetization [*Zhao et al.*, 1990]. This pole differs in declination only from the pole JG3, which was obtained from the same region [*Li et al.*, 1989]. Although apparent polar wander may have produced this difference, local block rotation or incorrect tilt correction or a combination of the both probably caused the difference. The pole JG 5 is reported to be from Permian turbidites [*Nie et al.*, 1989]. A primary magnetization is supported by a fold test. Four specimens at one site (out of seven site) show normal polarity, suggesting that the age of the magnetization is near the upper boundary of the Kiaman reversed interval at 250 Ma (S. Nie, personal communication, 1989). If the magnetization of the four normal polarity specimens is indeed primary, then the pole JG5 is probably the youngest among all poles plotted in Figure 8. The inclination predicted from the pole JG5 is consistent with that from the pole JG6 from the Urumqi region, which passes a fold test [*Sharps*, 1990], despite the difference in declination. The pole JG6 is indistinguishable from the pole JG4 of the underlying Qijiagou and Aoertu formations [*Sharps*, 1990], although the samples were collected from separate sections.

The important observation from our study is that despite the wide separation between the sampling localities, there is a good general agreement between inclinations from the southern and northwestern margins of the Junggar block, suggestive of coherence since Permian. In addition, the similarity between Permian inclination and present inclination indicates that the Junggar block has experienced little net latitude shift since Permian time. Unfortunately, concordant and exclusively negative inclinations also allow the possibility of remagnetization during the Kiaman reversed polarity interval. A post-Permian remagnetization, however, is unlikely because the magnetization age is constrained by a baked contact test, fold tests, and the data come from widely distributed regions and from different lithologies with an exclusively reversed magnetic polarity. In addition, the directions of magnetization from the late Paleozoic units are not resemble to any known younger magnetizations [*Y. A. Li et al.*, 1990; *Chen et al.*, 1991]. Second-order differences between results from the three regions are mainly in declination and are not readily explainable. The following are five potential explanations.

Apparent polar wander (APW). Magnetization ages from this study are constrained to be from early Carboniferous to late Permian. The relative ages of the magnetizations from different regions are not known. Within the same general sampling region, poles from different stratigraphic units are not consistent, indicating possible polar wander. For example, poles PG4 from the Qijiagou and Aoertu formations and the Permian strata are different from the pole from the late Permian pole PG5 of the Heavenly Lake area. In the Hoboksar region, the difference between the two pole positions could reflect apparent polar wander. In general, late Paleozoic APW for Junggar is reflected by a change of declination, from NNW to NNE (Figure 8).

Relative block rotation. At least part of difference in declination between results from different regions could be due to relative horizontal rotations about the vertical axis. This is suggested by the fact that the colatitudes predicted by these poles are very close to each other. Post-Permian thrusting which can cause local rotation has been well documented in the southern and northwestern margins of the Junggar block [*Avouac et al.*, 1990; *Feng et al.*, 1989; *Ulmishek*, 1986; *Lee*, 1985].

Uncertainty in tilt correction. The pole JP9 is based on a synfolding magnetization and thus paleo-horizontal reference may have been lost. Another pole (JP8) from the dikes intruding into the Liushugou Formation is based on an assumption that the dikes intruded into flat sedimentary beds. Any initial tilting of the sedimentary rocks could have produced a directional bias in the magnetization. For the Hoboksar region, results from plutons, dikes, and lava flows and remagnetized sedimentary units could be synfolding because volcanism, magmatism, and folding appear to be concurrent.

Secular variation (SV) of the geomagnetic field. It is difficult

TABLE 3. Hoboksar, Sedimentary, Lava Flow, Dike, Pluton (C_{2-3}-P_1, Latitude 47°10'N, Longitude 86°35'E)

Site	Age	N(M)	D	I	D_c	I_c	k	a_{95}	L	S(D)
Sedimentary Units (Honggulerun, Heshantou, and Sarlebulake Formations)										
8hd5*	D_{3h}	6(6)	138	35	154	-60	115	6.2	LS	25(103W)
8hd6*	D_{3h}	6(6)	125	48	126	-54	41	10.6	LS	25(103W)
8hd7*	D_{3h}	6(6)	134	42	141	-56	72	8	LS	25(103W)
8hc8*	C_{1h}	6(6)	286	-72	130	-42	109	6.4	LS	213(65W)
8hc9*	C_{1h}	6(6)	302	-50	125	-65	42	10.5	LS	213(65W)
8hc10*	C_{1h}	6(6)	308	-56	118	-59	71	8	LS	213(65W)
8hp16	C_{1s}	6(6)	160	-65	-	-	279	4	SS	70(75S)
8hp17	C_{1s}	6(6)	194	-69	-	-	242	4.3	SS	70(75S)
8hc19*	C_{1s}	6(6)	5	-52	154	-56	43	10.3	SS	260(70N)
8hp31	?	8(8)	184	-37	-	-	79	6.3	SS	223(80N)
Sites in Lava Flow, Dike, and Pluton										
8hd3*	?	6(6)	136	29	178	-60	17	16.7	PT	13(115W)
8hd4*	?	6(6)	133	35	166	-62	70	8.1	PT	13(115W)
8hp14	?	6(6)	181	-57	-	-	171	5.1	DK	-
8hp15*	?	5(5)	173	1	150	-47	349	4	FL	118(65S)
8hp18	?	6(6)	129	-57	-	-	20	28.6	PT	-
8hp20*	?	7(7)	148	-28	200	-60	37	10	PT	23(55E)
8hp21*	?	7(7)	131	-7	148	-57	53	8.4	FL	23(55E)
8hp22*	?	5(6)	145	-16	178	-56	334	4.2	FL	23(55E)
8hp23*g	?	6(6)	167	53	166	-32	-	2.5	FL	73(85S)
8hp24*	?	5(6)	150	17	133	-65	21	14.9	FL	73(85S)
8hp26*	?	7(7)	121	42	128	-29	63	7.7	FL	73(85S)
8hp27	?	6(6)	137	-52	-	-	173	5.1	DK	-
8hp29	?	7(7)	123	-64	-	-	201	4.3	DK	-
8hp30	?	5(5)	144	-63	-	-	322	4.3	DK	-
8hp32	?	3(6)	157	-61	-	-	49	17.9	DK	-
8hp36*	?	7(7)	325	-23	140	-77	-	5.1	PT	234(80N)
$Mean_b$	C_1-P	26(32)	148	-33	-	-	2	28.4	-	-
$Mean_a$	C_1-P	26(32)	-	-	151	-58	24	5.9	-	-
$Mean_b$*	C_1-P	17(17)	143	2	-	-	2	45	-	-
$Mean_a$*	C_1-P	17(17)	-	-	149	-57	23	7.7	-	-

Abbreviations are the same as in Table 1. For lithology, PT, pluton; LS, limestone; DK, dike; FL, lava flow; SS, sandstone. Mean direction of site 8hp23 is obtained from analysis of remagnetization circles. $Mean_b$* and $Mean_a$* are in situ and tilt-corrected mean directions of tilted sites only. $Mean_b$ and $Mean_a$ are in situ and tilt-corrected mean directions of all sites accepted.

*, Corrected for tilting.

to assess whether SV has been averaged in late Paleozoic paleomagnetic studies because of the presence of only a single magnetic polarity. However, for each pole in our study, the between-site k value is much smaller than the within-site k, suggesting that at least some secular variation has been averaged. The expected dispersion value (S) of Carboniferous-Permian is 17° at the equator [*Merrill and McElhinny*, 1983]. The mean S and its 95% confidence level for poles from Junggar is 17° ± 10°.

Permian back arc spreading. If Permian spreading occurred in the Junggar block as suggested by *Hsu* [1988], there may be a difference between the results from the northwestern (Hoboksar and Keramayi) and southern parts (Urumqi) of the Junggar block. Lack of precise age determination of magnetizations in each formation/unit does not allow us to test the back arc spreading hypothesis conclusively, although the hypothesis is permissible using the available data.

Because the discordance of the poles listed in Table 4 can be caused by different factors such as APW, SV, relative block rotation, and back arc spreading, we believe that a mean of the poles produces a conservative middle Carboniferous to late Permian paleomagnetic pole for Junggar. A Fisherian average of the six poles yields a pole at λ = 82°N, φ = 329°E, K =26 and A_{95} = 13.4°. The poles of JP8 and JP9 are excluded from the mean because of uncertainty in paleohorizontal control. Although the mean pole surely underestimates the APW during Carboniferous and Permian times, it reduces potential local effects and experimental errors associated with each individual pole and provides a general tectonic description for the whole Junggar block in late Carboniferous and Permian times.

In Figure 9, we compare late Paleozoic results from Junggar with late Paleozoic poles from Siberia, Kazakhstan, Tarim, and North China blocks. For Tarim, only three Carboniferous poles have been reported [*Bai et al.*, 1987; *Zhai et al.*, 1988; *Sharps*, 1990]. One is from the combined Lower and Middle Carboniferous Bashisuogong (C_{1b}), Bijingtawu (C_{2b}), and Suogedangtawa (C_{2s}) formations in the Wushi region of northwestern Tarim and is believed to represent a Permian overprint [*Sharps*, 1990]. The other two were both from the Upper Carboniferous Kangkeling Formation (Chinese system) in the Aksu region of northwestern margin of Tarim. The pole given by *Bai et al.* [1987] is also thought to represent a secondary magnetization, since the pole position is indistinguishable from those of the Permian. The pole given by *Zhai et al.* [1988] passes a fold test at p = 0.01. However, the latest folding phase in this region is Tertiary. The paleolatitude predicted by this pole is higher than that predicted by the Permian poles, whereas the general trend for Tarim during late Paleozoic (Devonian-Permian) is a northward motion [*Y. P. Li et al.*, 1990].

The Permian pole from the Tarim block is a mean of four independent studies from different rock types (limestone, sandstone, basalts and dikes) and widely separate locations [*Sharps et al.*, 1989] which has been confirmed by two recent studies [*Fang et al.*, 1989; Z. Meng, personal communication, 1989]. We have used this Permian pole for our comparison with Junggar.

It is clear from Figure 9 that the mean pole from the Junggar block is distinct from its surrounding blocks, suggesting substantial post-Permian relative motion. However, the late

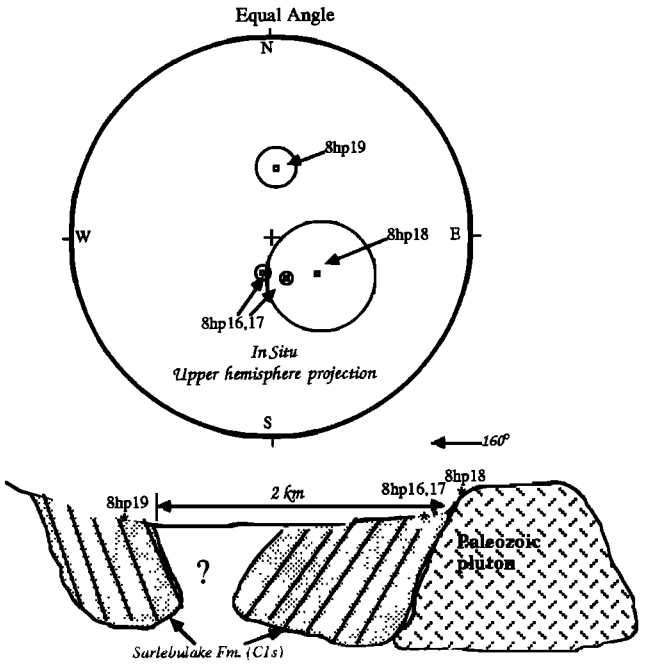


Fig. 6. Sketch of a baked contact test on the Sarlebulake Formation (C_{1s}) and the pluton intruding into the Formation. (Top): Equal-angle stereographic projections of site-mean directions plotted on upper hemispheres; (Bottom): Sketch of contact between the sandstones of the Sarlebulake Formation and the pluton intruding into the formation; asterisks are locations of sampling sites.

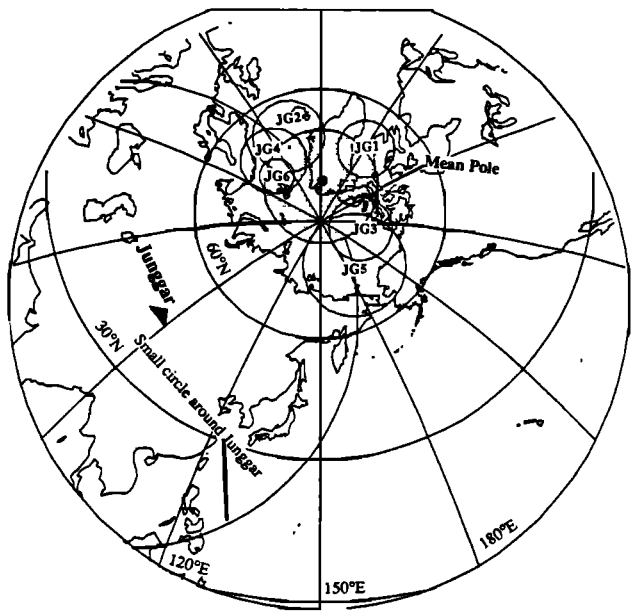


Fig. 8. Oblique polar projection of late Paleozoic paleomagnetic poles from the Junggar Block (Table 4). The mean pole is a shaded circle centered around a solid dot.

Carboniferous to Permian paleolatitude of the Junggar block predicted by our paleomagnetic results fit well with other geologic information.

Comparison of paleomagnetic poles also indicates late Carboniferous/Permian latitudinal proximity of Junggar, Tarim, Siberia, and Kazakhstan, supporting the hypothesis that (1) Junggar has been associated with Laurasia since at least middle Carboniferous [*Sharps*, 1990] and (2) Tarim arrived at the southern margin of Laurasia by Permian times. Middle-late Carboniferous flora in Junggar clearly belong to Angara, suggesting that Junggar had been attached or close to the south margin of the Siberia or Kazakhstan blocks since the middle Carboniferous times [*Hu*, 1985]. According to previous paleogeographic reconstructions based on flora/fauna and field geologic information, Junggar block was located between 35° and 50° latitudes during late Carboniferous [*Ziegler*, 1981; *Scotese*, 1984; *Watson et al.*, 1987; *Nie et al.*, 1990] . Our paleomagnetic results are in good agreement with those reconstructions.

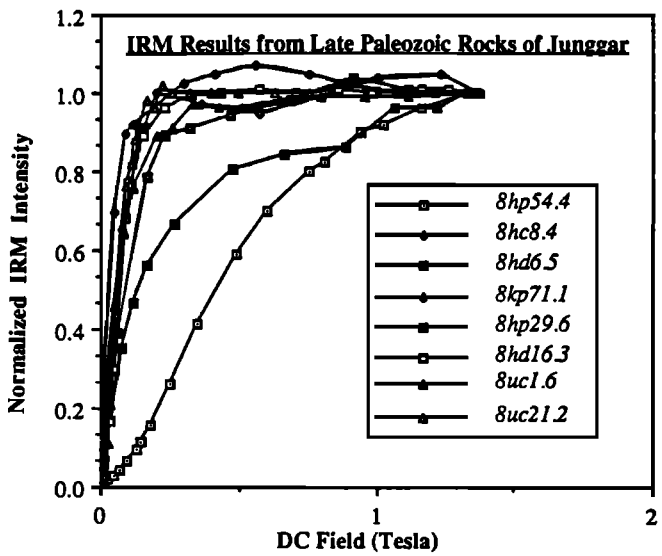


Fig. 7. Normalized isothermal remanent magnetization of specimens studied. Hoboksar region: 8hp54.4 is from a rhyolitic flow of the Karlargang Formation; 8hc8.4 is from limestone of the Heshantou Formation; 8hd6.5 is from limestone of the Honggulerun Formation; 8hp29.6 is from a dike; 8hd16.3 is from sandstone of the Sarlebulake Formation. Urumqi region: 8uc1.6 is from limestone of the Qijiagou Formation; 8uc21.2 is from sandstone of the Aoertu Formation. 8kp71.1 is from a late Paleozoic mafic dike in Keramayi [*Li et al.*, 1989].

There is a consensus among geologic community that the Tarim block accreted to the southern margin of Laurasia in late Carboniferous to early Permian [*Wang and Liu*, 1981; *Zhang et al.*, 1984; *Watson et al.*, 1987; *Li*, 1988; *Sharps*, 1990]. However, there has been a relative motion between the Junggar and Tarim blocks since then. For an example, the net latitudinal shift of Junggar since the Permian times is 4° (for a reference point at 45°N and 86.5°E, the change is from 41°N in the Permian to 45°N at present), whereas for the Tarim block the shift is 9° (31°N during the Permian to 40° at present for a reference point at 40°N, 80°E). The 5° difference or approximately 550 km probably represents post-collision continental shortening in the Tian Shan region (Figure 10). Continental shortening during the Cenozoic is evident in the Tian Shan Ranges [*Molnar and Tapponier*, 1975; *Avouac et al.*, 1990], and is estimated to be ~100 km across the Tian Shan near Urumqi based on a surface geology study [*Avouac et al.*, 1990]. We should emphasize that amount of the continental shortening based on paleomagnetic data is most likely. Given the nature of paleomagnetic data, any conclusion on this order (5°) is probably not statistically significant. Paleomagnetic data presented here also reveal the post-Paleozoic motion between Junggar and Kazakhstan, Siberia, and Tarim as evidenced by significantly different declinations predicted by the late Carboniferous/Permian paleomagnetic poles from Junggar, Siberian, Kazakhstan, and Tarim. The mean paleomagnetic directions from Junggar point basically to the north to northwest, whereas the expected directions of Tarim, Kazakhstan and Siberia point to from N40°E to even E10°S (Figures 9 and 11). Although this difference in declination could be accounted for by local structures, the results from three different, widely separate localities are so consistent that a relative rotation between the Junggar block and its adjacent units is most likely.

TABLE 4. Late Paleozoic Paleomagnetic Poles From Junggar, Western China

Location	Pole	Age	N	Long^P	Lat^P	Long^S	Lat^S	K	A_{95}	Reference
Bulungou	JG1	C_2	15	299	70	86.1	46.7	36	6 4	this study
Hoboksar	JG2	C_{2-3}-P	26	350	69	86.6	47 2	15	7 5	this study
Keramayi	JG3	C_3-P	23	230	80	84.7	45.4	16	7.8	*Li et al* [1989]
Qijiagou+Aocrtou	JG4	C_{2-3}	15	7	72	87.6	43.8	28	7.4	this study
Heavenly Lake	JG5	P_2	7	188	76	88.0	43.8	24	13	*Nie et al.* [1989]
Urumqi	JG6	P	78*	13	75	87.8	43.8	17	4	*Sharps* [1990]
Final Mean		C_{2-3}-P	6	329	82	-	-	26	13.4	this study
Liushugou†	JG7	C_2-P	4	231	69	87.8	43.8	46	13.7	this study
Dikes in Liushugou†	JG8	C_2-P	7	172	55	87.8	43.8	17	15.1	this study
West Junggar†	JG9	P(?)	11	178	51	84.2	45.5	58	6	*Zhao et al.* [1990]

N, Long^P, Lat^P, Long^S, Lat^S, K, A_{95} are number of site, pole position (longitude and latitude), sampling location (longitude and latitude), Fisher's precision parameter, and radius of the 95% confidence circle.

*, Number of sample.

†, Excluded from the final mean.

Post-Permian relative motion can be interpreted as clockwise rotations of Tarim, Kazakhstan, and Siberia relative to an arbitrarily fixed Junggar block. One possible consequence of the clockwise rotations is a left-lateral strike-slip motion between the Junggar block and its adjacent tectonic units, depending on locations of the Euler rotation poles. Paleomagnetic data from Tarim, when compared with the data from Siberia, suggest a significant eastward motion of Tarim relative to Siberia since Paleozoic [*Li*, 1988; *Zhao et al.*, 1990]. To accomplish this motion, a series of left-lateral strike-slip faults must exist between Tarim and Siberia. The left-lateral strike-slip motion along the boundary between the Junggar and Tarim blocks as suggested by paleomagnetic data from the Junggar block, is in good agreement with what is predicted from other independent paleomagnetic studies. It has been suggested that because of the penetration of India into the Asian continent, most parts of China have been pushed southeastward and rotated clockwise [*Tapponnier et al.*, 1982]. Limited Mesozoic paleomagnetic data from Tarim and Junggar indicate some 10° clockwise rotation of Tarim relative to Junggar since Jurassic [*Y. A. Li et al.*, 1990; *Chen et al.*, 1991]. Whether the left-lateral strike-slip motion is related to the collision of India and Asia remains to be confirmed by further late Mesozoic and Cenozoic paleomagnetic data from the Tarim and Junggar blocks.

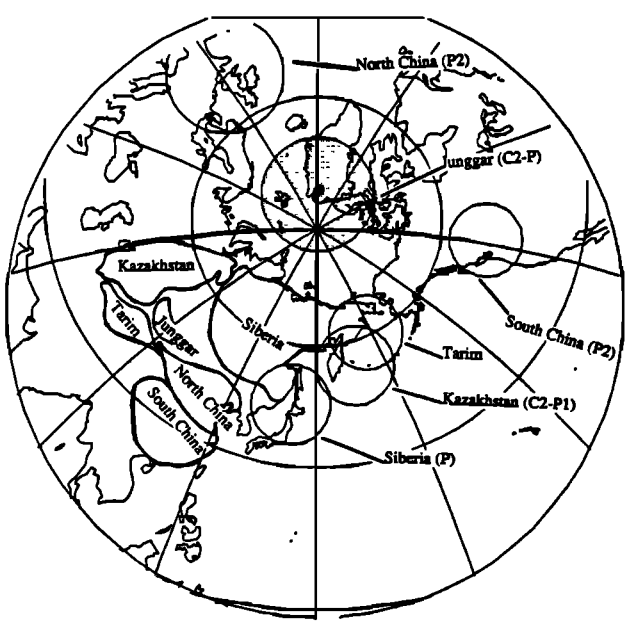


Fig. 9. Oblique polar projection (the same projection as Figure 8) of middle Carboniferous to Permian poles from Junggar (this paper), Tarim [*Sharps et al.*, 1989], Siberia [*Khramov et al.*, 1985], Kazakhstan [*Khramov et al.*, 1981], North and South China blocks [*Zhao and Coe*, 1987].

CONCLUSION

Carboniferous and Permian paleolatitudes obtained from widely separate localities around the Junggar block argue for its coherence since Permian time. Consistently steep inclinations, similar to the present one, indicate an affinity of the Junggar block with Laurasia during Permian [*Sharps*, 1990] and possibly Carboniferous as well, and little net shift in the north-south direction for the Junggar block since Permian. Prefolding magnetizations with an exclusively reversed magnetic polarity constrain the magnetizations to be between 250 and 320 Ma (the Kiaman reversed polarity interval). Whether concordant Carboniferous and Permian directions of magnetization suggest little APW or denote a remagnetization in the Kiaman reversed polarity interval (250-320 Ma) cannot be distinguished. In contrast, the Tarim block was located 10° south of the Junggar block during the Carboniferous and Permian [*Zhai et al.*, 1988; *Sharps et al.*, 1989] and was dominated by the Cathaysian flora, suggesting that Tarim was separated from Junggar by the ancestral Tian Shan. Since Permian times, some 550 km of continental shortening has occurred between the Tarim and Junggar blocks, presumably in

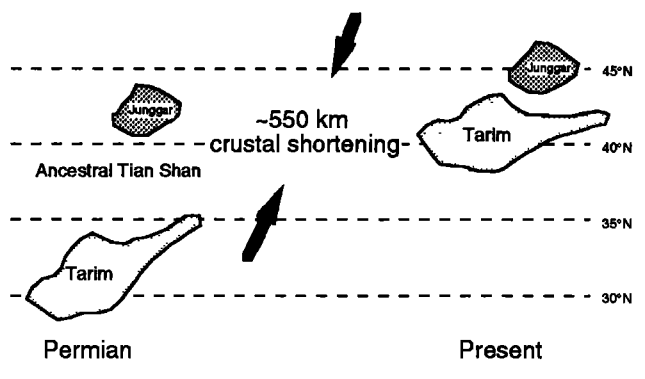


Fig. 10. Sketch of Permian reconstructions (constrained in latitude only) for the Tarim and Junggar blocks. Crustal shortening across the Tian Shan since the accretion of Tarim to the southern margin of Laurasia is estimated to be about 550 km based on paleomagnetic data (*Sharps et al* [1989] and this study).

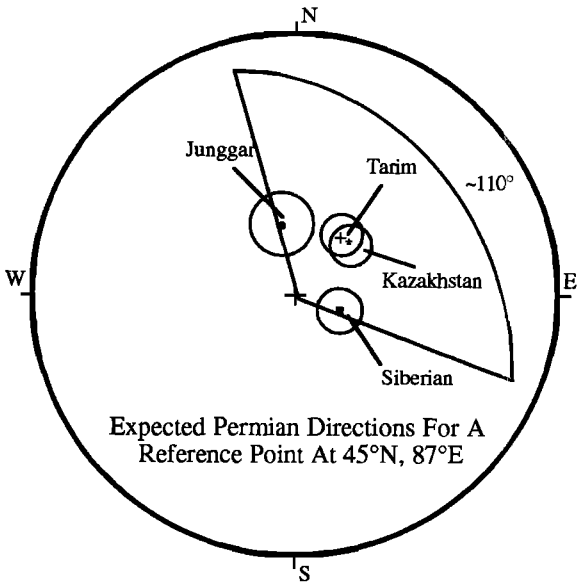


Fig. 11. Expected Permian directions from paleomagnetic poles of Junggar, Tarim, Kazakhstan, and Siberian for a reference point at 45°N, 87°E, plotted on equal-angle stereographic projection (lower hemisphere projection).

the Tian Shan region. Both North China and South China blocks are located in low latitudes during the Permian [*Zhao et al.*, 1987], and thus the collision between the North China and Junggar blocks must be post-Permian.

Finally, the discrepancy in declinations from Junggar, Tarim, Kazakhstan, and Siberian suggests post-Permian relative motions between Junggar and its adjacent blocks. A net result is that motion between Junggar and Tarim is likely to have a left-lateral strike-slip sense, consistent with other geological and paleomagnetic evidence.

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REFERENCES

Avouac, J. P., P. Tapponnier, M. Bai, H. You, and G. Wang, Field evidence for active faulting and folding along the northern piedmont of the Tien Shan (Xinjiang, China) (abstract), *Eos Trans. AGU, 71*, 1561, 1990.

Bai, Y., G. Chen, Q. Sun, Y. Sun, Y. Li, Y. Dong, and D. Sun, Late Paleozoic polar wander path for the Tarim Block and tectonic significance, *Tectonophysics, 139*, 145-153, 1987.

Bureau of Geology and Mineral Resources of Xinjiang , No.1 Regional Geological Surveying Party, List of geologic isotopic data from Xinjiang (in Chinese with English abstract), *Xinjiang Geol. 5*, 16-106, 1987.

Carroll, A., Y. Liang, S. Graham, X. Xiao, M. Hendrix, J. Chu, and C. McKnight, Junggar basin, northwest China: Trapped late Paleozoic ocean, *Tectonophysics, 181*, 1-14, 1990.

Chen, Y., et al., Paleomagnetic study of Mesozoic continental sediments along the northern Tien Shan (China) and heterogeneous strain in central Asia, *J. Geophys. Res., 96*, 4065-4082, 1991.

Coleman, R., Continental growth of northwest China, *Tectonics, 8*, 621-635, 1989.

Fang, D., G. Jin, H. Chen, Y. Guo, Z. Wang, S. Ying, and X. Tan, Discussion on paleomagnetism and tectonic evolution of Tarim plate (China) from late Paleozoic to Mesozoic, paper presented at the 4th National Conference on Paleomagnetism, Paleomagn. Comm. of Chin. Geol. Assoc. and Aerophys. Comm. of China, Hangzhou, 1989.

Feng, R. Coleman, G. Tilton, and X. Xiao, Tectonic evolution of the West Junggar Mountains, Xinjiang, China, *Tectonics, 8*, 1-22, 1989.

Fisher, R. A., Dispersion on a sphere, *Proc. R. Soc. London Ser. A, 217*, 295-305, 1953.

Harland, W. B., A. V. Cox, P. G. Llewelyn, A. G. Pickton, A. G. Smith, and R. Walter, *A Geologic Time Scale*, 131 pp., Cambridge University Press, New York, 1982.

Hsu, K. J., Relict back-arc basins: Principles of recognition and possible new examples from China, in *New Perspectives in Basin Analysis* edited by K.L. Kleinpell and C. Paola, pp. 245-263, Springer-Verlag, New York, 1988.

Hu, Y., Characteristics of the Permian floras in the western part of China (in Chinese), *Reg. Geol. China, 12*, 100-108, 1985.

Huang, T. K., J. Ren, C. Jiang, Z. Zhang, and D. Qin, *Geotectonic Evolution of China*, 203 pp., Springer-Verlag, New York, 1987.

Khramov, A. N., G. N. Petrova, and D. M. Pechersky, Paleomagnetism of the Soviet Union, in *Paleoreconstruction of the Continents, Geodyn. Ser.*, vol. 2, edited by M. W. McElhinny and D. A. Valencio, pp. 159-194, AGU, Washington, D. C., 1981.

Khramov, A. N., E. L. Gurevich, R. A. Komissarova, E. P. Osipova, S. A. Pisarevsky, V. P. Rodionov, and I. P. Slautsitais, Paleomagnetism, microplates and Siberian plate consolidation, *J. Geodyn., 2*, 127-139, 1985.

Kirschvink, J. L., The least-squares line and plane and the analysis of paleomagnetic data, *Geophys. J. R. Astron. Soc., 62*, 699-718, 1980.

Kwon, S.-T., G. R. Tilton, R. G. Coleman, and Y. Feng, Isotopic studies bearing on the tectonics of the West Junggar Region, Xinjiang, China, *Tectonics, 8*, 719-727, 1989.

Lee, K. Y., Geology of the petroleum and coal deposits in the Junggar basin, Xinjiang Uygur Zizhiqu, northwest China, *US Geol. Surv. Open File Rep, 85-230*, 53 pp., 1985.

Li, Y. A. Li, Y., R. Sharps, M. McWilliams, Q. Li, W. Zhang, Mesozoic paleomagnetic results from the Junggar block, Xinjiang Province, northwest China (abstract), *Eos Trans. AGU, 71*, 1292, 1990.

Li, Y. P., Paleomagnetism of western China and the southern Sierra Nevada, Ph.D. dissertation, 234 pp., Stanford Univ., Stanford, Calif., 1988.

Li, Y., R. Sharps, M. McWilliams, N. Amos, Y. A. Li, Q. Li, and W. Zhang, Paleomagnetic results from late Paleozoic dikes from northwestern Junggar block, northwestern China, *Earth Planet. Sci. Lett., 94*, 123-130, 1989.

Li, Y. P., M. McWilliams, R. Sharps, A. Cox, Y. A. Li, Q. Li, Z. Gao, Z. Zhang, and Y. Zhai, A Devonian paleomagnetic pole from red beds of the Tarim block China, *J. Geophys. Res., 95*, 19,185-19,198, 1990.

Liao, Z., L. Lu, N. Jiang, F. Xia, F. Sun, Y. Zhou, S. Li, and Z. Zhang, Carboniferous and Permian in the western part of the east Tianshan Mountains, *Guide Book Excursion 4*, 39 pp., 11th Int. Congr. of Carboniferous Stratigraphy and Geol., Beijing, 1987.

Ma, X., Lithospheric dynamics map of China and adjacent seas, scale 1:4,000,000, Geol. Publ. House, Beijing, China, 1987.

McElhinny, M., Statistical significance of the fold test in paleomagnetism, *Geophys. J. R. Astron. Soc., 8*, 338-340, 1964.

McFadden, P., and D. Jones, The fold test in paleomagnetism, *Geophys. J. R. Astron. Soc., 67*, 53-58, 1981.

Merrill, R., and M. McElhinny, *The Earth's Magnetic Field: Its History, Origin and Planetary Perspective*, 401 pp., Academic, San Diego, Calif., 1983.

Molnar, P., and P. Tapponnier, Cenozoic tectonics of Asia: Effects of a continental collision, *Science, 189*, 419-425, 1975.

Nie, S., D. B. Rowley, R. Van Der Voo, and M. Li, Late Paleozoic Paleomagnetic results from Northwest China and Permian Asian Plate Reconstructions (abstract), *Eos Trans. AGU, 70*, 1071, 1989.

Nie, S., D. B. Rowley, and A. Ziegler, Constraints on the locations of Asian microcontinents in Palaeo-Tethys during the Late Paleozoic,

in *Paleozoic Palaeogeography and Biogeography*, vol. 12, edited by W. S. McKerrow and C. R. Scotese, pp. 397-408, *Mem. Geol. Soc. London* , 1990.

Scotese, C. R., An Introduction to this volume: Paleozoic paleomagnetism and assembly of Pangea, in *Plate Reconstruction From Paleozoic Paleomagnetism, Geodyn. Ser.* vol. 12, edited by R. Van der Voo, C. R. Scotese, and N. Bonhommet, pp.1-10, AGU, Washington, D. C., 1984.

Sharps, R., Paleomagnetism and tectonics of northwest China, Ph. D. dissertation, 225 pp., Stanford Univ., Stanford, Calif., 1990.

Sharps, R., M. McWilliams, Y. P. Li, A. Cox, Z. Zhang, Y. Zhai, Z. Gao, Y. A. Li, and Q. Li, Lower Permian paleomagnetism of the Tarim Block, northwestern China, *Earth Planet. Sci. Lett., 92*, 275-291, 1989.

Smith, A. B., Late Paleozoic biogeography of East Asia and palaeontological constraints on plate tectonic reconstructions, *Philos. Trans. R. Soc. London, Ser. A 326*, 189-227, 1988.

Tapponnier, P., A. Peltzer, A. Dain, R. Armijo, and P. Cobbold, Propagating extrusion tectonics in Asia: New insights from simple experiments with plasticine, *Geology, 10*, 611-616, 1982.

Ulmishek, G., Geology and petroleum resources of basins in western China, *Argonne Natl. Lab. Rep., ANL/ES-146*, 131 pp., 1986.

Wang, H., and B. Liu, Paleogeographic frame of China since Middle Proterozoic (in Chinese), *J. Stratigr., 5*, 77-89, 1981.

Watson, M. P., A. B. Hayward, D. N. Parkinson, and Z. M. Zhang, Plate tectonic history, basin development and petroleum source rock deposition onshore China, *Mar. Pet. Geol., 4*, 205-225, 1987.

Xinjiang Stratigraphic Compiling Group, *Regional Stratigraphic Tables of Northwest China, Volume on Xinjiang* (in Chinese), 496 pp., Geol. Publ. House, Beijing, 1981.

Zhai, Y., Z. Zhang, Y. A. Li, Q. Li, Y. P. Li, M. McWilliams, A. Cox, and R. Sharps, A study of Upper Carboniferous paleomagnetism for the Tarim Block (in Chinese with English abstract), *Geoscience, 2*, 43-56, 1988.

Zhang, Z. M., J. G. Liou, and R. G. Coleman, An outline of the plate tectonics of China, *Geol. Soc. Am. Bull., 95*, 295-312, 1984.

Zhao, X., and R. S. Coe, Paleomagnetic constraints on the collision and rotation of north and south China, *Nature, 327*, 141-144, 1987.

Zhao, X., R. Coe, Y. Zhou, H. Wu, and J. Wang, New paleomagnetic results from northern China: collision and suturing with Siberia and Kazakhstan, *Tectonophysics, 181*, 43-81, 1990.

Ziegler, A. M., Paleozoic paleogeography, in *Paleoreconstruction of the continents, Geodyn. Ser.*, vol. 2, edited by M. W. McElhinny and D. A. Valencio, pp. 31-37, AGU, Washington, D. C., 1981.

Zijderveld, J. D. A., AC demagnetization of rocks: Analysis of results, in *Methods in Paleomagnetism*, edited by D. W. Collinson, K. M. Creer, and S. K. Runcorn, pp. 254-286, Elsevier, New York, 1967.

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